

instead rely on external vendors for fixes and timelines. This not only slows innovation but also introduces uncertainty in mission-critical applications.

The challenge becomes more acute in areas requiring customisation and security assurance. Without ownership of the underlying IP, tailoring systems to specific operational environments, such as Indian road conditions, defence needs, or industrial constraints, is severely constrained. At the same time, the inability to audit internal system behaviour raises serious concerns about hidden vulnerabilities, backdoors, and supply chain risks, particularly in strategic sectors.

As a result, many systems operate as opaque 'black boxes': inputs and outputs are visible, but internal logic remains inaccessible. This lack of visibility leads to a broader loss of control, in which external stakeholders effectively dictate key decisions regarding system behaviour, updates, and lifecycle management. Over time, this undermines technological self-reliance and exposes critical infrastructure to risks that cannot be fully mitigated domestically.

In essence, despite building and operating these systems, we do not truly own them. We interact with them, depend on them, and deploy them at scale, yet core control lies elsewhere. The result is a stark reality: we have built and integrated systems at scale, yet ownership of core intelligence often lies outside our borders. We remain participants, almost 'guests', inside systems that are critical to our own infrastructure.

Why now?

While India's dependence on external technologies is not new, its strategic urgency has intensified significantly in recent years. This shift is not incidental; it stems from converging forces across policy, global supply chains, and security realities

that have collectively exposed the risks of technological dependence.

A key driver has been the government's strong policy push since 2014. Flagship initiatives such as Make in India and Atmanirbhar Bharat have laid the groundwork for domestic manufacturing and self-reliance. Building on this, targeted missions, including the National Semiconductor Mission, National AI Mission, and National Quantum Mission, aim to address critical gaps in high-technology domains. These are not merely economic programmes; they reflect a deliberate push towards end-to-end technological sovereignty, where India can design, develop, manufacture, and control its own technology stack.

Simultaneously, global supply chain disruptions have acted as a wake-up call. Recent events have exposed the fragility of semiconductor ecosystems, where production is concentrated in a few regions and firms. This concentration has led to shortages, delays, and cost escalations across industries, from automotive to consumer electronics, while highlighting the risks of overdependence on a limited set of global players. For India, this creates vulnerability to geopolitical shifts, trade restrictions, and unforeseen disruptions.

In parallel, security concerns have grown more pronounced. Modern electronic systems are deeply layered, with potential vulnerabilities spanning hardware design to firmware implementation. Risks such as hardware-level backdoors, firmware exploits, and unverified IP blocks are particularly critical in sectors like defence, infrastructure, and transportation. Without full control over the technology stack, detecting, preventing, and responding to such threats becomes extremely challenging.

Together, these factors have transformed what was once a long-term developmental goal into an immediate strategic priority. The

focus has shifted from cost and convenience to control, resilience, and security, making indigenous capability development not just desirable but essential.

India's semiconductor growth story, spanning automotive, industrial, IoT, and defence sectors, is no longer just an economic opportunity; it is now central to national resilience, strategic autonomy, and long-term technological control.

Gaps in the current ecosystem

Despite notable progress in policy, funding, and ecosystem development, India's semiconductor and electronics landscape continues to face several structural gaps that hinder its journey towards true self-reliance. These challenges are not isolated; they are deeply interconnected and must be addressed holistically.

- **Fragmented development**, driven by siloed teams and limited cross-functional coordination, reduces interoperability and significantly complicates integration, debugging, and scaling.

- **A critical skill gap in hardware-software integration** remains a major challenge. There is a shortage of engineers skilled in both hardware and software, with most specialists focused on one area and lacking experience in firmware and system-level design. This hinders the development of efficient, deeply integrated systems.

- **Continued reliance on foreign EDA tools** is another major concern. These tools are essential for chip design, yet India depends heavily on a small number of global providers. This dependence limits flexibility, raises costs, and constrains long-term technological autonomy.

- **A lack of emphasis on early-stage validation** also weakens the ecosystem. Systems often fail not because of flawed design, but because of inadequate testing. Delayed checks for EMI, thermal performance, and real-world